

Hypothermia with Trauma



Section I: Scenario Demographics

Scenario Title:	Hypothermia
Date of Development:	(05/05/2014)
Target Learning Group:	☐ Juniors (PGY 1 – 2) ☒ Seniors (PGY ≥ 3) ☐ All Groups

Section II: Scenario Developers

Scenario Developer(s):	Dr. Stephen Miller
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Section III: Curriculum Integration

Learning Goals & Objectives	
Educational Goal:	To consider possible pathologies in patients presenting hypothermia.
CRM Objectives:	To communicate clearly To effectively manage a team – ask for help early, delegate workload To avoid fixation error
Medical Objectives:	Identify and prioritize treatment of various injuries in a hypothermic patient Concurrently implement multiple active and passive rewarming therapies

Case Summary: Brief Summary of Case Progression and Major Events
A 30 year-old female is brought into the ED at 4 AM by a man who found her lying at the side of the road with no coat or shoes. It is minus 30 degrees Celsius outside. On arrival she has a reduced LOC, labored breathing, a right-sided pneumothorax, cyanotic extremities, a left radius & ulna fracture, and a right tib-fib fracture. The team is required to use both active and passive rewarming strategies. Regardless of the team's efforts, the patient in this case will arrest. Upon ROSC, they are required to continue rewarming as well as to address the other traumatic injuries.

References
Polderman KH (2009) Mechanism of action, physiological effects, and complications of hypothermia. Crit Care Med. 37 (7suppl): S186-202
Marx, J. A., Hockberger, R. S., Walls, R. M., & Adams, J. (2013). <i>Rosen's emergency medicine: Concepts and clinical practice</i> . St. Louis: Mosby.

Hypothermia with Trauma



Section IV: Scenario Script

A. Clinical Vignette: To Read Aloud at Beginning of Case

An approximately 30 year old female is brought into the ED at 4 AM by a man who found her lying at the side of the road. It is minus 30 degrees Celsius outside and she has no coat or shoes. The man does not know her and is unable to provide any additional history except that she was blue and having trouble breathing when he found her. She is noted to have a decreased LOC and laboured breathing. She has obvious deformities of her left forearm and right leg.

B. Scenario Cast & Realism

Patient:	☒ Computerized Mannequin	Realism: <i>Select most important dimension(s)</i>	☒ Conceptual
	☒ Mannequin		☒ Physical
	☐ Standardized Patient		☐ Emotional/Experiential
	☐ Hybrid		☐ Other:
	☐ Task Trainer		☐ N/A
Confederates	Brief Description of Role		
Nurse	Available to help only if asked		

C. Required Monitors

☒ EKG Leads/Wires	☒ Temperature Probe	☐ Central Venous Line
☒ NIBP Cuff	☒ Defibrillator Pads	☐ Capnography
☒ Pulse Oximeter	☐ Arterial Line	☒ Rectal Temperature Probe

D. Required Equipment

☒ Gloves	☒ Nasal Prongs	☒ Scalpel
☒ Stethoscope	☒ Venturi Mask	☐ Tube Thoracostomy Kit
☒ Defibrillator	☒ Non-Rebreather Mask	☐ Cricothyroidotomy Kit
☒ IV Bags/Lines	☒ Bag Valve Mask	☒ Chest Tube Tray
☒ IV Push Medications	☒ Laryngoscope	☐ Central Line Kit
☐ PO Tabs	☐ Video Assisted Laryngoscope	☒ Bladder Catheter
☐ Blood Products	☒ ET Tubes	☒ Rapid Infuser
☐ Intraosseous Set-up	☐ LMA	☒ Bair Hugger

E. Moulage

Female mannequin dressed in jeans and long sleeve shirt (not winter clothes). Left forearm and right lower leg are deformed due to fracture.

F. Approximate Timing

Set-Up:	10 min	Scenario:	10 min	Debriefing:	20 min
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Hypothermia with Trauma



Section V: Patient Data and Baseline State

A. Patient Profile and History			
Patient Name: Unknown		Age: ~30	Weight: 61kg
Gender: ☐ M ☒ F		Code Status: Unknown	
Chief Complaint: Altered LOC			
History of Presenting Illness: Found at the side of the road on a cold night with no coat, no shoes, cold, cyanotic and having difficulty breathing.			
Past Medical History:	Unknown	Medications:	Unknown
Allergies: Unknown			
Social History: Unknown			
Family History: Unknown			
Review of Systems:	CNS:	Agitated and confused	
	HEENT:	Unknown – patient can't answer questions	
	CVS:	Unknown – patient can't answer questions	
	RESP:	Unknown – patient can't answer questions	
	GI:	Unknown – patient can't answer questions	
	GU:	Unknown – patient can't answer questions	
	MSK:	Unknown – patient can't answer questions	
B. Baseline Simulator State and Physical Exam			
☒ No Monitor Display		☐ Monitor On, no data displayed	☐ Monitor on Standard Display
HR: 55/min	BP: 98/60 mmHg	RR: 16/min	O ₂ Sat: ?%
Rhythm: Sinus brady	T: 30°C	Glucose: 5.1 mmol/L	GCS: 8 (E 2 V 2 M 4)
General Status: Critically ill, decreasing LOC, cold			
CNS:	Withdraws, opens eyes, and moans in response to painful stimuli		
HEENT:	Pupils sluggish and 4mm reactive		
CVS:	No murmur		
RESP:	Decreased air entry to right lung fields. Crepitus and bruising along right lateral chest.		
ABDO:	Soft, non-tender. No external bruising		
GU:	No blood noted.		
MSK:	Obvious deformity of right forearm and right lower leg, both closed	INT:	Extremities cyanosed

Hypothermia with Trauma

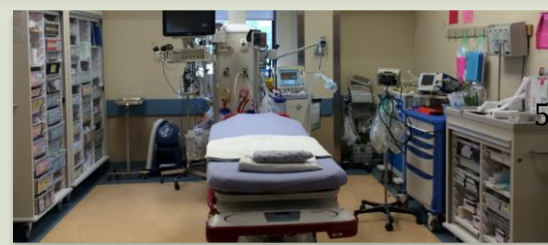


Section VI: Scenario Progression

Scenario States, Modifiers and Triggers			
Patient State	Patient Status	Learner Actions, Modifiers & Triggers to Move to Next State	
1. Baseline State Rhythm: Sinus brady HR: 55/min BP: 98/60 mmHg RR: 16/min O ₂ Sat: ?% T: 30°C GCS: 8 (E 2, V 2, M 4)	Cold, GCS 8, and increased work of breathing.	<u>Learner Actions</u> ☐ Full set of vitals, monitors, O ₂ ☐ Insert rectal temp probe ☐ 2 large-bore IVs & bolus 1L NS ☐ Warm IV fluids ☐ Check cap sugar (5.1) ☐ Labs including tox, troponin, group and screen ± TSH ☐ Rewarming with Bair hugger ☐ ± Insert foley for CBI with warmed fluids ☐ Full Px to identify injuries ☐ Order CXR or perform E-FAST ☐ ECG	<u>Modifiers</u> <i>Changes to patient condition based on learner action</i> <u>Triggers</u> <i>For progression to next state</i> - 3 minutes → 2. Worsening #1
2. Worsening #1 HR→48/min BP→90/60 mmHg RR→14/min O ₂ Sat→85% T→29°C	Unchanged	<u>Learner Actions</u> ☐ CXR or E-FAST (if not yet) ☐ Initiate Bair hugger, foley, warm fluids if not yet done ☐ Consider needle decompression R chest ☐ Prepare for R chest tube ☐ Prepare for intubation	<u>Modifiers</u> - O ₂ applied → sat to 90% <u>Triggers</u> - 2 minutes → 3. Worsening #2
3. Worsening #2 HR→40/min BP→84/52 mmHg RR→20/min O ₂ Sat→87% T→29°C	Unchanged	<u>Learner Actions</u> ☐ Insert R chest tube ± two chest tubes to L for warm fluids ☐ Performs EFAST (if not yet) ☐ Re-checks cap sugar (2.1) , gives glucose ☐ ± Intubation with warmed air through ETT	<u>Modifiers</u> <u>Triggers</u> - 2 minutes → 4. Cardiac arrest
4. Cardiac arrest Rhythm→VF T→28°C	Unconscious	<u>Learner Actions</u> ☐ Performs effective CPR ☐ Immediate defibrillation ☐ Consider 1 dose of epi ☐ Consider giving NaHCO ₃ ☐ Re-check cap sugar (2.1) , give glucose (if not done prior) ☐ Intubation	<u>Modifiers</u> - Defibrillation: to PEA rhythm <u>Triggers</u> - 2 rounds CPR + defibrillation x1 → 5. ROSC
5. ROSC Rhythm→Sinus + PVCs HR→90/min BP→110/70 mmHg RR→ 12 (intubated) O ₂ Sat→90%	Colour improving	<u>Learner Actions</u> ☐ Call ICU ± CV surgery for bypass to rewarm ☐ Use warmed air through ETT ☐ Post-intubation CXR ☐ Xray injured limbs & splint	<u>Modifiers</u> <u>Triggers</u> END CASE



Hypothermia with Trauma



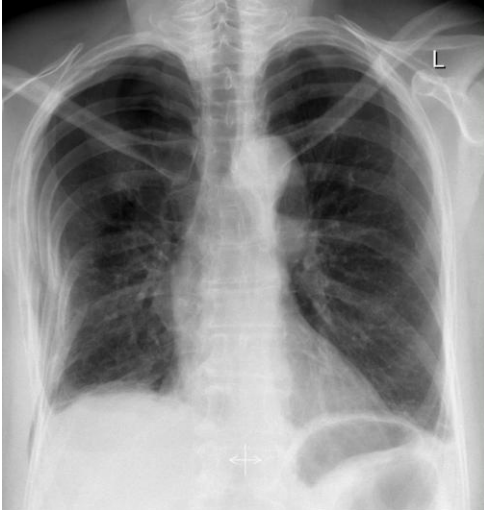
Section VII: Supporting Documents, Laboratory Results, & Multimedia

Laboratory Results

No lab results given during case.

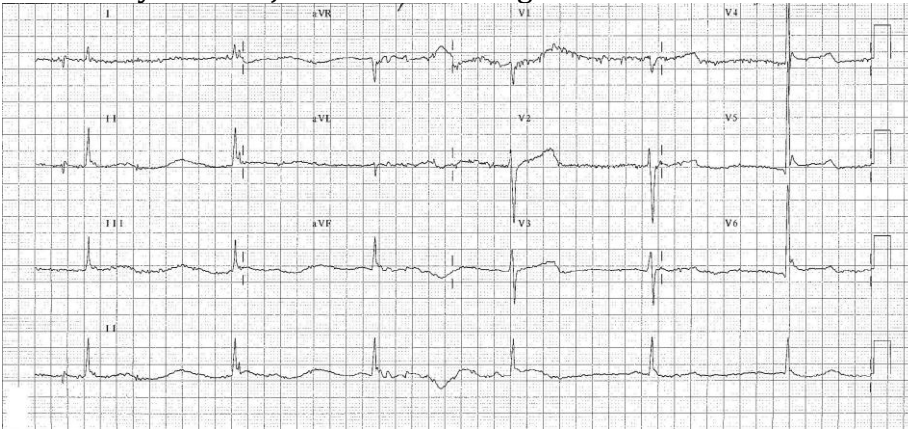
Images (ECGs, CXRs, etc.)

CXR: pre-intubation



(Source: <http://radiopaedia.org/cases/pneumothorax-due-to-rib-fractures-1>)

ECG: bradycardia + J waves + shivering artifact



(Source: <http://cdn.lifeinthefastlane.com/wp-content/uploads/2011/03/hypothermia-shiver-artefact.jpg>)

Ultrasound Video Files

FAST – No FF RUQ

FAST – Normal pericardium

Lung US – R lung no lung sliding

Lung US – L lung no lung sliding



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Section VIII: Debriefing Guide

General Debriefing Plan	
☐ Individual	☒ Group
☐ With Video	☒ Without Video
Objectives	
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Sample Questions for Debriefing	
1. How do you feel your team worked together during this case? Did everyone feel like they had clearly delineated roles? 2. When did you realize that the patient also had traumatic injuries? How did it feel to manage two complex presentations at once? 3. What are possible rewarming strategies? Were they all used or considered in this case? 4. Do you have any strategies to avoid fixation on one symptom (i.e. fixation on hypothermia while missing pneumothorax)?	
Key Moments	
Recognition that hypothermia is highest priority pathology	
Initiation of multiple (active and passive) rewarming techniques	
Identification and treatment of pneumothorax	
Cardiopulmonary resuscitation	

